ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework

Sherwin Stephen Chan[1,*], J-Anne Yow[1,2,*], Yi Heng San[1], Vasanthamaran Ravichandram[1], Yifan Wang[1], Lek Syn Lim[1], Wei Tech Ang[1,2]
[*]These authors contributed equally

Abstract—**Robot-assisted feeding has the potential to enhance the independence of individuals requiring assistance, yet the bite transfer process remains particularly challenging, especially for those with complex conditions. In this paper, we present ORBiT, a novel Real2Sim2Real framework designed to optimize bite transfer in robot-assisted feeding. By integrating motion capture-driven, high-fidelity soft-body simulation with systematic parameter tuning, ORBiT effectively replicates realistic head, neck and jaw dynamics during feeding interactions to provide a safe simulation-driven approach to optimize bite transfer strategies. In our approach, motion capture data drives a personalized dynamic head model that, together with a comprehensive parameter search over variables such as entry angle, exit angle, exit depth, height offset, and distance to mouth, identifies the bite transfer parameters that minimize contact forces on the user. The optimal parameters are then transferred to a real-world robotic system and validated through a pilot user study involving five subjects. Results from real user evaluations mirror the trends in simulation, indicating that bite transfer parameters, especially those related to entry and exit angles, substantially affect user comfort and overall satisfaction. Our findings validate that simulation-derived optimizations can effectively guide improvements in bite transfer strategies, laying the groundwork for a safe, personalized approach to robot-assisted feeding. Supplementary videos can be found at: https://youtu.be/a2pklEIAkOA.**

Index Terms—**Robot-Assisted Feeding, Physical Human-Robot Interaction, Physically Assistive Devices, Simulation and Animation, Real2Sim and Sim2Real**

I. INTRODUCTION

Assisted feeding is a critical need for individuals with limited physical abilities due to conditions such as multiple sclerosis, and spinal cord injuries, which can lead to restricted head and neck mobility, spasms, and masseter spasticity. Despite notable progress in robot-assisted feeding, widespread adoption remains limited. A major challenge lies in the close physical contact required at sensitive regions like the mouth, a challenge compounded for care recipients with complex conditions such as limited mouth opening or involuntary spasms. Directly testing bite transfer strategies — delivering food from a utensil into a user's mouth — in these high-risk scenarios raises significant safety concerns, hindering the development of robot feeding systems that are both safe and effective.

Recent advances in human-in-the-loop (HITL) simulation have proven valuable for safety-critical assistive tasks, such as assisted dressing [1] and exoskeleton control [2], and have demonstrated successful simulation-to-real-world (Sim2Real) transfer [3]. However, simulating physical human-robot interaction (p-HRI) for robot-assisted feeding remains under-explored, largely due to the difficulty of accurately modeling the head's soft-body dynamics—critical for capturing mouth–spoon contact forces. While prior HITL simulations typically rely on rigid-body representations, recent work [4] has introduced static soft-body head models that capture essential contact dynamics for users with minimal neck mobility. However, these models are not capable of replicating the complex dynamics of users with active head and neck movements, resulting in less accurate simulation of bite transfer dynamics. This limitation underscores the need for a high-fidelity, dynamic simulation framework that accurately captures the nuanced interactions during bite transfer for realistic human-robot interaction.

In this paper, we introduce ORBiT (Optimizing Robot-assisted Bite Transfer), a novel Real2Sim2Real framework that integrates motion capture driven, advanced soft-body simulation to safely enhance bite transfer in robot-assisted feeding. Our approach begins by capturing motion data from a subject to drive a dynamic, soft-body head simulation, thereby reproducing realistic human movements that are difficult to replicate with scripted trajectories (Real2Sim). We create a personalized human model that combines rigid and soft-body dynamics, and parameterize the bite transfer trajectory into key variables, including distance to the mouth, entry angle, exit depth, exit angle, and height offset. A systematic parameter search in simulation is then used to identify the set of parameters that minimizes contact forces on the human model. The optimal parameters derived from simulation are transferred to a real-world system for user studies, closing the Sim2Real loop.

Notably, although our parameter set was optimized using data from a single subject, it outperforms conventional fixed 90° entry and exit angles used in existing approaches [5]. Our user study — the first of its kind to evaluate bite transfer

The research was conducted at the Future Health Technologies at the Singapore-ETH Centre, which was established collaboratively between ETH Zurich and the National Research Foundation Singapore. This research is supported by the National Research Foundation Singapore (NRF) under its Campus for Research Excellence and Technological Enterprise (CREATE) programme.

[1]School of Mechanical and Aerospace Engineering, Nanyang Technological University Singapore
[2]Singapore-ETH Centre, Future Health Technologies Programme
Corresponding authors' e-mails: {sherwins001, janne001}@e.ntu.edu.sg

parameters with real users — reveals that variables such as entry and exit angles and exit depth considerably impact user comfort, safety, and satisfaction. The trends observed in our user study closely matched those found in simulation, validating our framework. These findings underscore the importance of personalizing bite transfer strategies and demonstrate that our Real2Sim2Real framework provides a robust foundation for further improvements in robot-assisted feeding by enabling the simulation of realistic user conditions, such as spasms, limited mobility, and impaired coordination, common among many care recipients.

Overall, our findings indicate that our Real2Sim2Real framework provides a promising starting point for identifying effective bite transfer parameters, and they underscore the critical role of personalization in accommodating the variability in bite transfer dynamics among individuals. The key contributions of this work are:

- **ORBiT - A Real2Sim2Real Framework with Dynamic Human Simulation:** We present a novel pipeline that integrates motion capture driven, high-fidelity soft-body simulation with systematic parameter tuning. This personalisable simulation environment features a dynamic soft-body human head model with an articulated mandible, enabling effective Sim2Real transfer for bite transfer optimization.
- **Pilot User Study:** We conduct the first user study evaluating bite transfer parameters, revealing the key factors that critically influence user comfort, safety, satisfaction, and ease of use.

II. Related Work

A. Bite Transfer in Robot-Assisted Feeding

Early autonomous bite transfer systems primarily relied on mouth pose detection to track the user's mouth [6], [7]. They typically stop at a predetermined position from the user's mouth, operating under the assumption that the user can lean forward to complete the bite, which may be difficult for users with limited upper neck mobility. This limitation has motivated recent research into in-mouth bite transfer. Approaches such as force-compliant controllers [8] and methods that utilize visual and haptic feedback for adaptive controller selection [5] have been developed to enhance comfort and safety.

A common assumption in the current methods is that a fixed, preplanned trajectory for bite transfer with force-compliant controllers is sufficient for user comfort. Although such controllers can adapt to unexpected contact forces and correct for minor deviations, they fail to capture the full complexity of bite transfer dynamics. For example, the fixed 90° entry and exit angle used in [5] were found to be uncomfortable in our user studies. To understand how bite transfer parameters affect user comfort, our previous work [4] introduced a soft-body head model in simulation to study these parameters systematically. However, it did not incorporate dynamic neck actuation and lacked real-user validation.

B. Physical Human-Robot Interaction Simulation for Assisted Feeding

Realistic p-HRI simulation is necessary for simulating physical human-robot interaction for assistive tasks. In this work, we define realism as the accurate modeling of the human shape, movement and physics. While numerous biomechanical and robotics simulators exist [9]–[14], few integrate the fidelity required for accurately simulating physical human-robot interaction (p-HRI) in assistive feeding tasks. Prior work introduced platforms such as Assistive Gym [15] and RCareWorld [16] to unify p-HRI simulation across assistive tasks. However, these systems face significant limitations. Assistive Gym's human model lacks the soft-body dynamics necessary for capturing realistic contact interactions, and it oversimplifies critical anatomical features, such as lacking an actuated jaw and mouth cavity for utensil insertion, and simplified neck movements. Similarly, while RCareWorld advances the integration of human-robot interaction in assistive contexts by improving soft-body dynamics, its human actuation relies on predefined configurations rather than real motion capture data, limiting the realism of simulated movements.

To the best of our knowledge, ORBiT is the first Real2Sim2Real framework to leverage a realistic, personalized human model driven by real motion capture data to systematically explore optimal bite transfer strategies and evaluate their impact on users based on comfort, safety, satisfaction, and ease of use. This work showcases the potential of simulation-driven optimization for enhancing the performance of robot-assisted feeding.

III. Approach

ORBiT starts by capturing a subject's motion data to drive a dynamic, soft-body head simulation, ensuring realistic human head and neck movements. We then parameterize the bite transfer trajectory using key variables and systematically search for the optimal set that minimizes contact forces on the digital human. Finally, these optimal parameters are transferred to a real-world system for a pilot user study involving 5 subjects with no mobility limitations.

A. Motion Capture Data Acquisition

We capture motion data from one of the authors using a Qualisys motion capture system (AB, Sweden). Retro-reflective markers were placed on the subject and the spoon (Fig. 2). For safety, the spoon was mounted on a utensil mount with a breakaway mechanism [17] and grasped by a robotic arm (xArm6, UFactory, China). Marker trajectories and robot data were recorded synchronously at 200Hz.

To capture the reference motion, we first calibrated the robot arm to hold the spoon such that the spoon tip was touching the subject's lower lip. Next, the robot arm was retracted by 2 cm using a position controller, simulating a scenario where the spoon is positioned 2 cm away from the user's mouth along the sagittal plane. The subject then leaned forward to take a bite and moved back to their starting position. We repeated

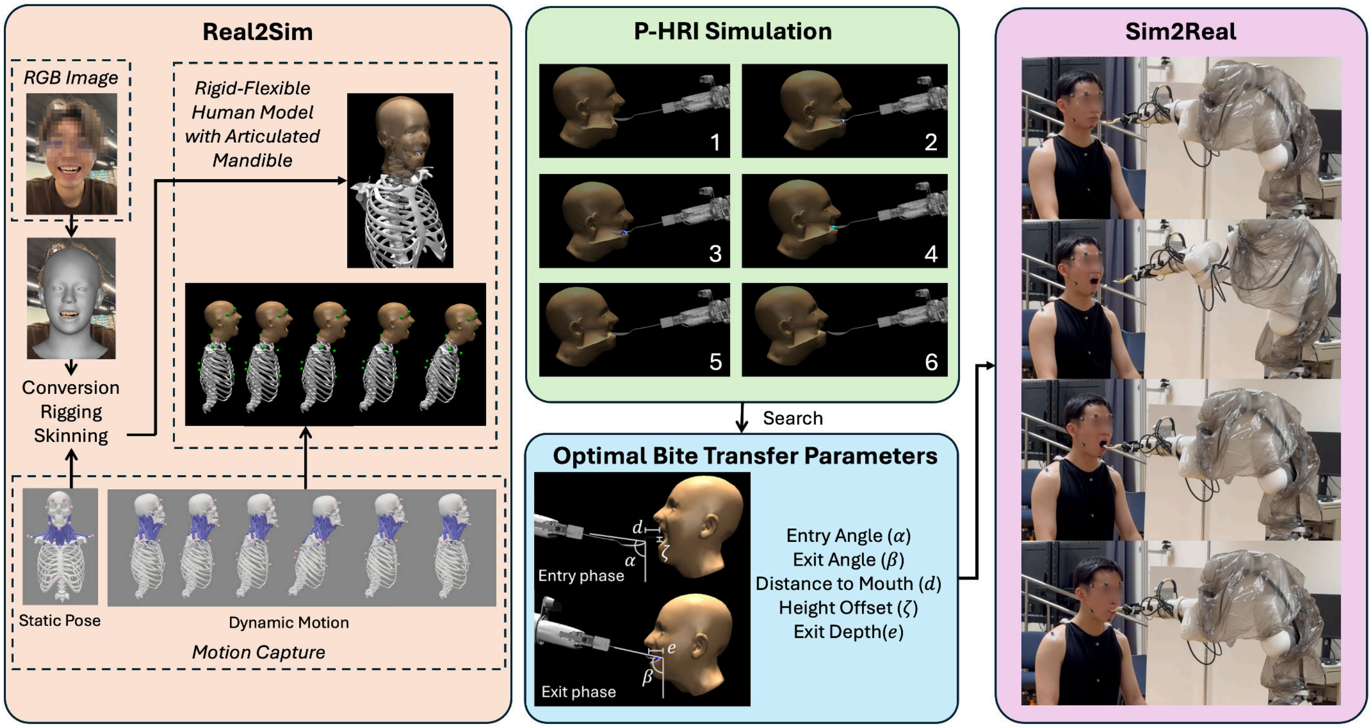

Fig. 1: **ORBiT**: A Real2Sim2Real framework to safely optimize robot-assisted bite transfer. (Left) A personalized flexible head model is generated from RGB image through surface mesh reconstruction. It is converted, rigged and skinned with a personalised rigid head and neck model created using OpenSim to produce a rigid-flexible human model with an articulated mandible and 33 degrees of freedom in MuJoCo. (Top Center) p-HRI simulation with the human model and a robotic arm grasping a spoon. Human motion is driven by real motion capture data of the subject while the robot trajectory is systematically varied to search for the optimal bite transfer policies. (Bottom Center) Bite transfer parameters used in the search include entry angle α (80–100° in 10° steps), exit angle β (80–120° in 10° steps), distance to mouth d (2–8cm in 2cm steps), height offset ζ (+1cm, 0cm, -1cm) and exit depth e (0– 5cm in 1cm steps). (Right) Optimal bite transfer parameters identified in simulation were transferred to real-world settings (Sim2Real) and validated through pilot user trials.

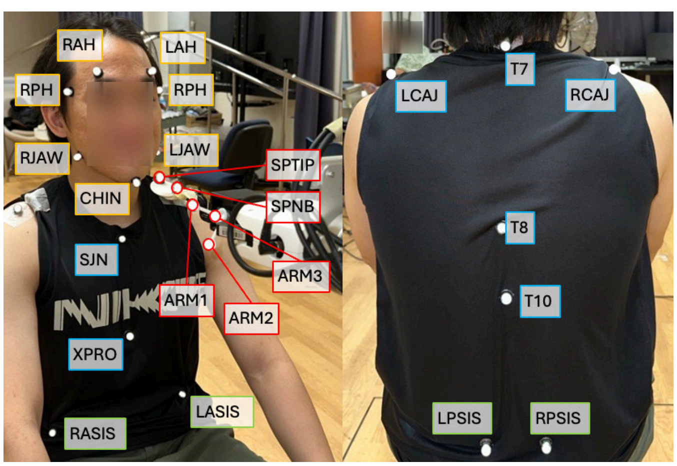

Fig. 2: Motion capture marker placements: Yellow-labeled markers are placed on the head, blue-labeled markers on the torso, green-labeled markers on the pelvis, and red-labeled markers on the robot arm and spoon.

each motion twice, repeating it for distances of 4cm, 6cm and 8cm.

B. Simulation Framework

1) Generation of a Moving Flexible Head Model: We create a personalized human head model for simulation by enhancing the OpenSim head and neck model [18] with a custom marker set and a manually implemented 6-DoF jaw joint to enable mandible articulation. Using OpenSim's scaling tool, we adjust the model and specify the subject's mass before converting it to MuJoCo with an open-source converter [19], [20]. The final 33-DoF model — 6 DoF at the root, 3 DoF at each cervical vertebra (C7 to C1), and 6 DoF at the jaw — allows replication of subtle head and neck movements associated with feeding.

We use DECA [21] to reconstruct a personalized 3D mesh from an RGB image and follow [4] to create a simplified tetrahedral mesh for soft-body dynamics in MuJoCo. Based on human tissue properties [22], [23], we set Young's modulus to 5×10^4, the Poisson Ratio to 0.4 and assign a total mass of 5.7 kg evenly across vertices, with overlapping skeletal structures (skull to C7) set near zero to avoid redundancy.

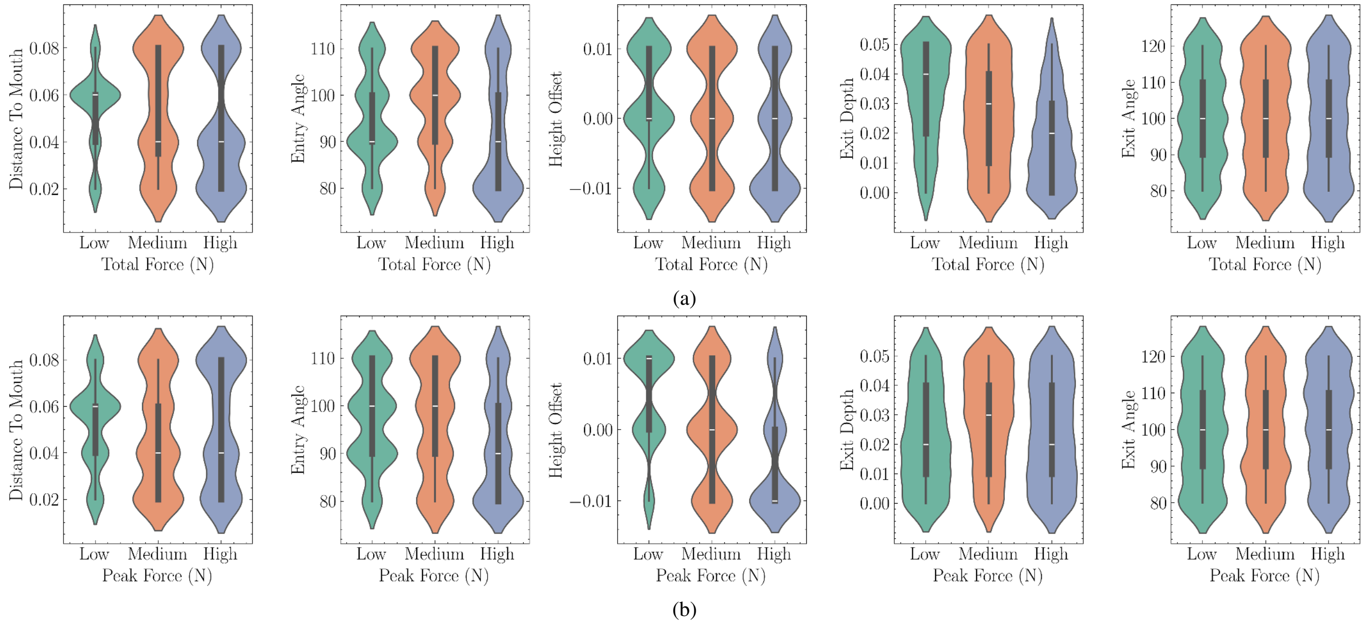

Fig. 3: Violin plots illustrating the distributions of bite transfer parameters categorized into low, medium, and high force groups for (a) total contact forces (N) and (b) peak force (N). A distance to mouth of approximately 0.06 m is predominantly associated with lower forces, while a lower entry angle tends to correspond with higher forces. Increased height offset values generally coincide with lower peak forces. Larger exit depth are linked with higher total contact forces but lower peak force.

MuJoCo's `tendon` element was used to model skin-skeletal interaction in [4], but some motions with rapid acceleration introduced latency and simulation stability issues. To address this, we adopted a rigging approach that establishes a hierarchical parent-child relationship between skeletal components and the flexible mesh. We used the `pin` element to rigidly connect key contact points and the `equality connect` attribute to constrain vertices, thereby enhancing stability in dynamic bite transfer simulations.

2) Bite Transfer Parameters: To analyse the physical human-robot interaction forces exerted on the user by a spoon during bite transfer, we parameterize bite transfer based on the distance to mouth, entry angle, height offset, exit depth and exit angle (Fig. 1). A comprehensive parameter search is conducted within the simulation environment to identify the optimal set of bite transfer parameters that minimizes contact forces on the human model. We compare simulation results across a static head model and a dynamically moving head model to quantify the benefits of dynamic motion in achieving safer and more comfortable bite transfer.

For the dynamic head model, where the simulated head moves forward to bite the spoon, we use MuJoCo's `mocap body` feature to drive the soft-body head based on motion capture data from a real user. This approach enables realistic human motion that is difficult to achieve with randomized or scripted movements, allowing us to obtain more realistic physical contact information between the spoon and mouth.

C. Simulation Results

Fig. 3 presents the distribution of contact forces for the dynamic head model across all parameter combinations and Table I shows the parameter importance. The total contact force is the sum of the average contact force per time step throughout the bite transfer, an indicator of overall comfort during the bite transfer; the peak force is the maximum instantaneous contact force, reflecting any instantaneous discomfort.

Both analyses highlight exit depth, entry angle, and exit angle as the most influential factors affecting peak force and total contact force. Exit depth was the most critical factor for total force, with larger exit depth corresponding to lower total forces. This reduction can be attributed to the user and the robot pulling away together, shortening the duration of spoon contact. With a synchronized withdrawal, sustained contacts are minimized.

Entry angle has high importance relative to peak force. A low entry angle (80°) results in higher forces due to the spoon tip hitting the roof of the mouth, whereas a 90° entry angle results in lower peak and total contact forces. While exit angle ranks high in feature importance, the median exit angle remains similar across force groups, suggesting that its combination with other factors have a higher impact on contact forces.

Height offset and distance to mouth rank lower in importance for peak and total contact force. A 6 cm distance to mouth tends to yield lower peak and total contact forces, suggesting optimal positioning for that particular subject. A higher height offset helps reduce peak force, as lower offsets

TABLE I: Random Forest Parameter Importances for Two Force Metrics. Higher values indicate greater relative importance in predicting the specified force metric.

Parameter	Random Forest Importance	
	Total Contact Forces	Peak Force
Exit Depth	0.305	0.339
Entry Angle	0.245	0.402
Exit Angle	0.199	0.209
Height Offset	0.190	0.036
Distance to Mouth	0.061	0.013

increase the risk of lip collisions or unintended pressure.

The results suggest that an optimal bite transfer strategy should prioritize adjusting entry and exit angles while managing spoon exit depth for a smooth, controlled retraction. The most comfortable bite transfer is achieved with a relatively neutral 90°–100° entry angle, a higher exit angle of 90°–110°, and a high exit depth of 3–5cm. These settings minimize peak and total contact forces, ensuring a gentle, controlled interaction for improved comfort and safety.

IV. User Study on Bite Transfer

The optimal parameters derived from simulation are transferred to a real-world robot-assisted feeding system, as shown in Table II. We conducted a pilot study involving five subjects without mobility limitations (all male, ages 18-50) to evaluate the effectiveness of these parameters in practice. This study was approved by the Nanyang Technological University Institutional Review Board (IRB-2023-571).

A. Procedure

The experimental setup was consistent with that described in Section III-A, with two key additions: (1) detection of the subject's mouth pose using 3D Dense Face Alignment (3DDFA-v2) [24], [25], and (2) the use of an admittance controller for participant safety during all trials.

After obtaining informed consent, participants provided demographic information and completed pre-study questionnaires. Motion capture markers were then placed based on Fig. 2 to capture their natural movements for subsequent analysis. Participants were briefed on the bite transfer parameters and study procedure before being provided with cereal balls to place onto the spoon for each trial.

A familiarization session was conducted first, during which the robot arm moved towards the subject at a 90° entry angle without retracting. This allowed participants to acclimate to the system. Participants then completed five experimental conditions, each testing different bite transfer parameter settings. For each condition, bite transfers were performed at four distances (2, 4, 6, 8 cm), with each distance repeated twice, resulting in a total of 40 bite transfers per subject. The conditions experienced were as follows:

1) **Initial Self-Selection** ***(Self-Init)***: Participants specified their own preferred bite transfer parameters, assessing how accurately new users can identify their optimal settings.

TABLE II: User Study Parameters for Each Condition, Including User-Selected Parameters.

Condition	Entry Angle (°)	Height Offset (m)	Exit Angle (°)	Exit Depth (m)
Static-Sim	110	0.01	110	0.01
Dyn-Sim	90	0.00	110	0.05
Fix-90	90	0.00	90	0.05
Self-Init-1	95	0.01	110	0.03
Self-Final-1	110	0.01	110	0.05
Self-Init-2	90	0.01	95	0.01
Self-Final-2	100	0.01	110	0.05
Self-Init-3	90	0.00	100	0.00
Self-Final-3	90	0.00	100	0.03
Self-Init-4	95	0.01	100	0.03
Self-Final-4	92	0.01	105	0.03
Self-Init-5	105	0.00	105	0.05
Self-Final-5	105	0.00	105	0.05

2) **Static Simulation** ***(Static-Sim)***: Bite transfers were executed using the optimal parameters derived from simulation with a static head model.
3) **Dynamic Simulation** ***(Dyn-Sim)***: Bite transfers utilized parameters derived from simulation where the head dynamically moved based on motion capture data during the transfer.
4) **Fixed 90°** ***(Fix-90)***: A standard fixed trajectory with 90° entry and exit angles was employed, serving as a baseline for comparison with existing methods [5].
5) **Final Self-Selection** ***(Self-Final)***: Participants once again selected their own bite transfer parameters, allowing us to examine any evolution in user preferences after experiencing the other conditions.

To examine changes in user preferences after experiencing the system, *Self-Init* was always performed first, and *Self-Final* was always performed last. The orders of *Static-Sim*, *Dyn-Sim* and *Fix-90* were counterbalanced across subjects to mitigate order effects. The parameter values used are shown in Table II.

After each set, subjects completed a questionnaire using a 5-point Likert scale to rate their satisfaction, perceived safety, comfort, ease of bite transfer, and the naturalness of the bite transfer. At the end of the trial, subjects were asked to rank all experiment conditions from the most preferred to least preferred. Additionally, they provided qualitative feedback on the reasons behind their top and bottom choices.

Our user study was designed to address the following research questions:

- **RQ1**: Do bite transfer parameters influence user comfort during bite transfer?
- **RQ2**: Can optimal bite transfer parameters derived from the dynamic motion of one user be generalized to other users?
- **RQ3**: How accurately can new users of robot-assisted feeding systems identify their own optimal bite transfer settings based on their subjective preferences?

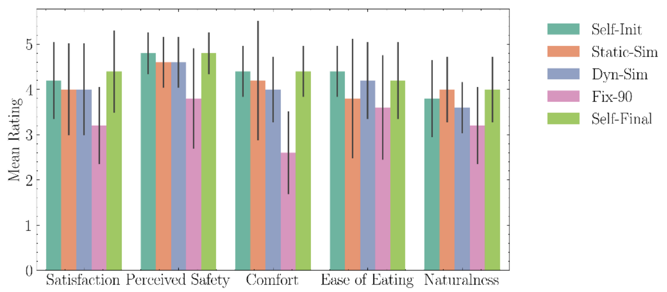

Fig. 4: Bar charts illustrating the aggregated ratings of participants for each scenario. In general, participants preferred their modified bite transfer parameters the most, followed by their initial parameters.

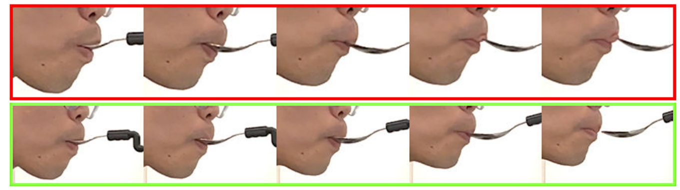

Fig. 5: Image stills comparing two exit angles during spoon withdrawal. (Top) A 90° entry and exit angle, where the spoon presses against the user's top lip as it exits, causing discomfort. (Bottom) A 90° entry angle with a 110° exit angle, allowing the spoon to follow the mouth's contour more naturally and improving overall comfort. Exit depth and height offset remain consistent between both conditions.

- **RQ4**: How effectively do force-based metrics serve as proxies for user comfort during bite transfer?

B. *Results and Discussion*

Fig. 4 summarizes the subjective evaluations of various bite transfer scenarios, revealing distinct ratings in comfort, satisfaction and perceived safety across conditions, validating the need for personalized bite transfer parameters. Additionally, the user preference rankings indicate the following order from most to least preferred: *Self-Final*, *Self-Init*, *Dyn-Sim*, *Static-Sim*, with *Fix-90* being the least preferred.

1) Influence of Bite Transfer Parameters on User Experience: Notably, the *Fix-90* condition received the lowest ratings across most metrics, suggesting that a fixed 90° entry and exit angle leads to an unfavorable bite transfer experience (**RQ1**). Video analysis and user feedback indicate that discomfort mainly stemmed from the spoon's exit angle pressing against the upper lip (Fig. 5).

In contrast, the *Self-Final* condition, where subjects adjusted their settings after experiencing other conditions, obtained highest ratings, indicating that users refined their preferences with increased interaction and feedback. As shown in Table II, most subjects preferred a higher exit angle ($\geq$100°), which video footage shows to follow the spoon's natural curvature during withdrawal and reduces friction. Entry angles exhibited greater variability (90°–105°), further emphasizing the need for personalization. Subjects also increased their preferred exit depth to 3–5 cm, while longer bite transfer distances (e.g., 8 cm) were deemed less comfortable, due to the additional effort required. These trends observed in real-world settings align closely with our simulation findings, validating our Real2Sim2Real framework.

Dyn-Sim parameters generally ranked as the next best alternative after self-selected settings, with *Static-Sim* parameters following closely. This suggests that optimal parameters derived from a single user's dynamic simulation can serve as a promising starting point for others, though further fine-tuning is required (**RQ2**).

While most subjects favored their own final selections, one participant preferred the *Dyn-Sim* condition over his self-selected settings, a promising indication that our simulation-derived parameters can, in some cases, outperform user-selected parameters and that users may not always accurately identify their optimal parameters (**RQ3**). Overall, final preferences shifted after exposure to various strategies, demonstrating that users may need time and experience to refine their choices. Furthermore, all subjects in this study had a technical background, which likely aided their ability to understand and refine bite transfer parameters. For care recipients, who may not have the same level of familiarity with such adjustments, identifying suitable parameters could be even more challenging, underscoring the need for a Real2Sim2Real bite transfer optimization to provide a strong initial parameter set, particularly for care recipients who may struggle with manual fine-tuning.

2) Force Metrics as Predictors of User Comfort: Interestingly, lower contact forces did not consistently correlate with higher user preference (**RQ4**). This discrepancy may be due to several factors. First, the overall forces were modest ($\leq$ 8 N), so minor variations might not be perceptible to users. Second, force readings for some subjects increased over time, which may indicate that as users became more comfortable with the system, they exhibited less restraint during interaction, resulting in slightly higher forces. Third, the force-torque sensor on the spoon captures only overall interaction forces at the mounting point, missing localized pressure distributions on the soft tissues of the mouth. In contrast, our soft-body simulation captures tissue deformation and the full contact area, offering a more comprehensive picture of the forces experienced during bite transfer.

Additionally, arm motion during bite transfer also played a role in user comfort. One reason some participants rated *Dyn-Sim* lower was the high variation in entry and exit angles (ranging from 90° to 110°), leading to large arm movements, particularly evident in one of the robot's joints. These jarring movements may have contributed to perceived discomfort and reduced safety, also noted in [8], [26]. This suggests that optimizing both utensil motion and robotic arm kinematics to achieve smoother, more subtle movements is crucial for enhancing the overall user experience.

Overall, our user study highlights the importance of personalizing bite transfer strategies and validates the parameter trends identified in our simulation. Even among subjects without mobility limitations, who are inherently more adaptable to the spoon, the benefits of personalized bite transfer parameters were evident in terms of improved comfort, safety, and satisfaction. These findings demonstrate that the parameters derived from ORBiT using our Real2Sim2Real framework not only provide a robust starting point for optimization but also highlight the critical need to safely study and refine bite transfer strategies, especially for care recipients with mobility limitations.

V. Conclusion and Future Work

In this work, we introduced ORBiT, a novel Real2Sim2Real framework for optimizing bite transfer in robot-assisted feeding. By integrating motion capture-driven, high-fidelity soft-body simulation with systematic parameter tuning, our approach captures realistic head and neck dynamics to enable robust Sim2Real transfer of bite transfer parameters. Our pilot user study demonstrated that personalized settings, especially those related to entry and exit angles, substantially improve user comfort, safety, and satisfaction. Notably, even parameters derived from a single user's dynamic simulation provide a strong baseline, although further personalization is needed to accommodate individual variability.

Overall, our framework offers a safe, simulation-driven environment for exploring and refining bite transfer strategies before deployment, laying a solid foundation for future applications with care recipients facing complex conditions such as spasms and limited mobility. Future work will focus on validating our framework across diverse user conditions, refining robotic arm kinematics to further enhance both safety and performance, and conducting evaluations with more representative users of robot-assisted feeding.

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